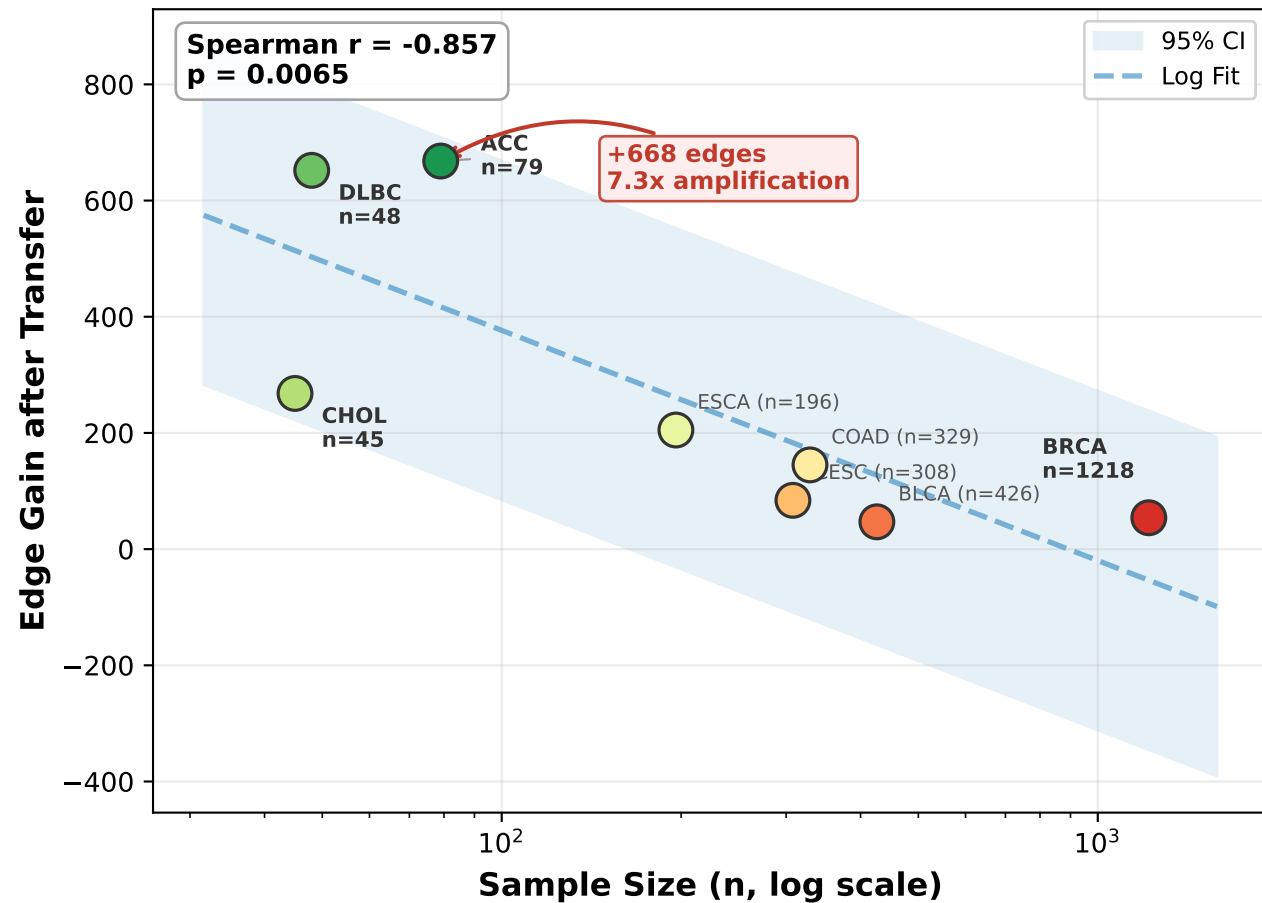
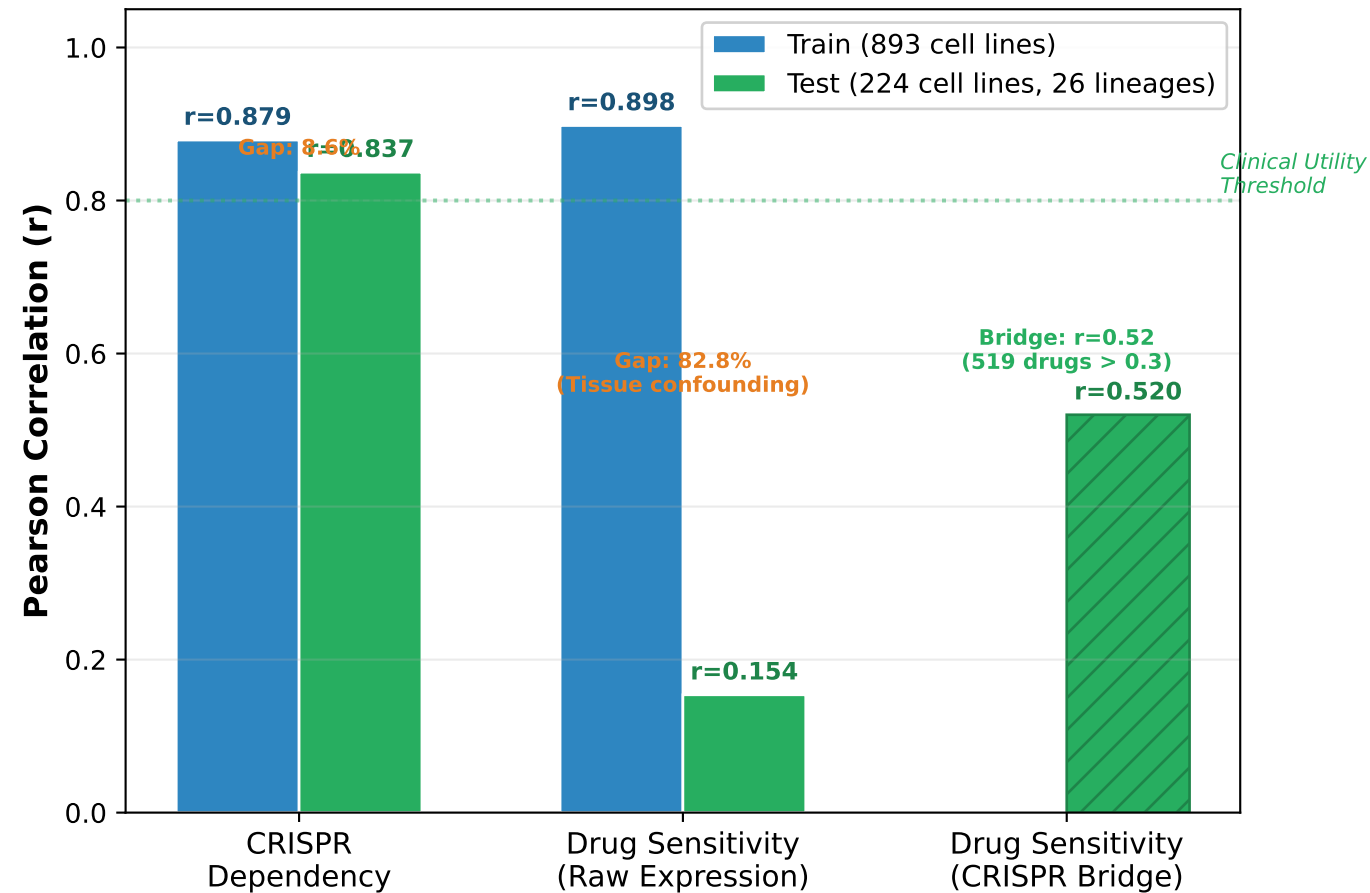


Zero-Shot Transfer: Small-Sample Amplification



Prospective Validation: 80/20 Stratified Split



DepMap pre-training + TCGA fine-tuning. CRISPR bridge uses causal features to bypass tissue-type confounding in drug prediction.